

Food Delivery Management System

Data Flow Diagrams (DFD)

This document presents the Data Flow Diagrams (DFD) for the Food Delivery Management System using standard notation: **rounded rectangles** for processes, **rectangles** for external entities, and **open rectangles** for data stores. It follows top-down decomposition — from the system-wide Context Diagram (Level 0), through the major functional processes (Level 1), down to a detailed breakdown of Order Management (Level 2).

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#	Diagram	Description
1	Context Diagram (Level 0)	System as a single process with external entities
2	Level 1 DFD	Major processes and data stores
3	Level 2 DFD	Detailed breakdown of Order Management (3.0)

DFD Notation Used

Symbol	Meaning
Double circle, e.g. ((3.0 Order Management))	Process — transforms incoming data flows into outgoing ones
Rectangle, e.g. [Customer]	External Entity — a source or destination of data outside the system
Open-ended rectangle, e.g. D3 Order DB	Data Store — where data is held at rest
Labelled arrow	Data Flow — the movement of data between entities, processes, and stores

1. Context Diagram (Level 0)

The highest-level DFD, showing the entire Food Delivery Management System as a single process interacting with its five external entities.

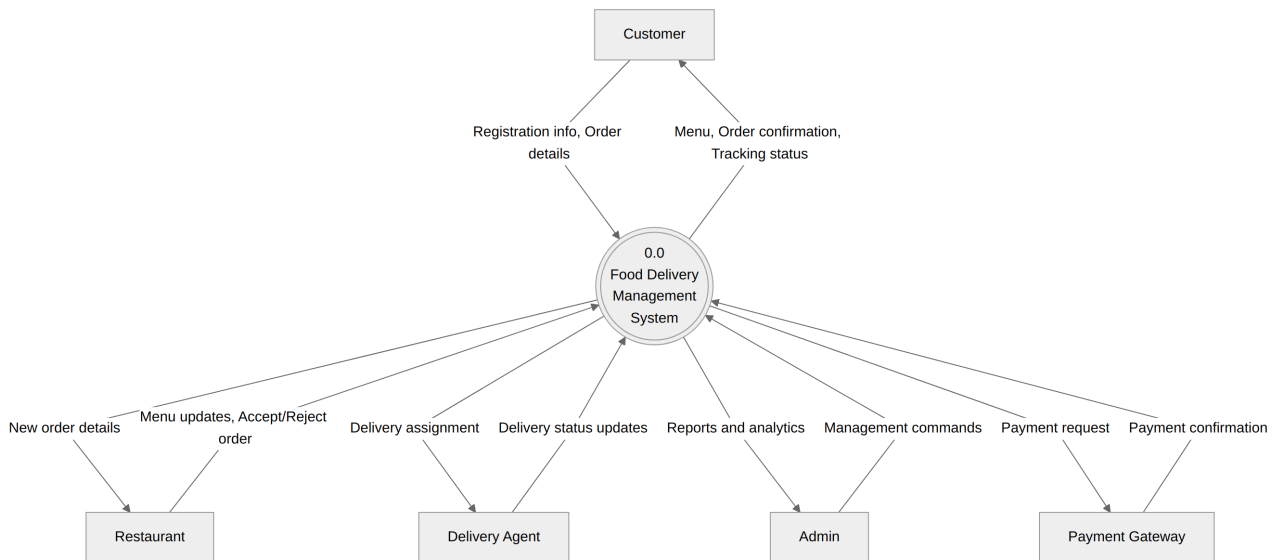


Figure 1: Context Diagram (Level 0)

Key Points

- The system exchanges registration, order, menu and tracking data with the **Customer**.
- It exchanges menu updates and order accept/reject decisions with the **Restaurant**.
- It exchanges delivery assignments and status updates with the **Delivery Agent**.
- It exchanges management commands and reports with the **Admin**.
- It exchanges payment requests and confirmations with the external **Payment Gateway**.

2. Level 1 DFD

Decomposes the system into six major processes, each reading from and writing to its own data store, showing how data moves between entities, processes, and storage.

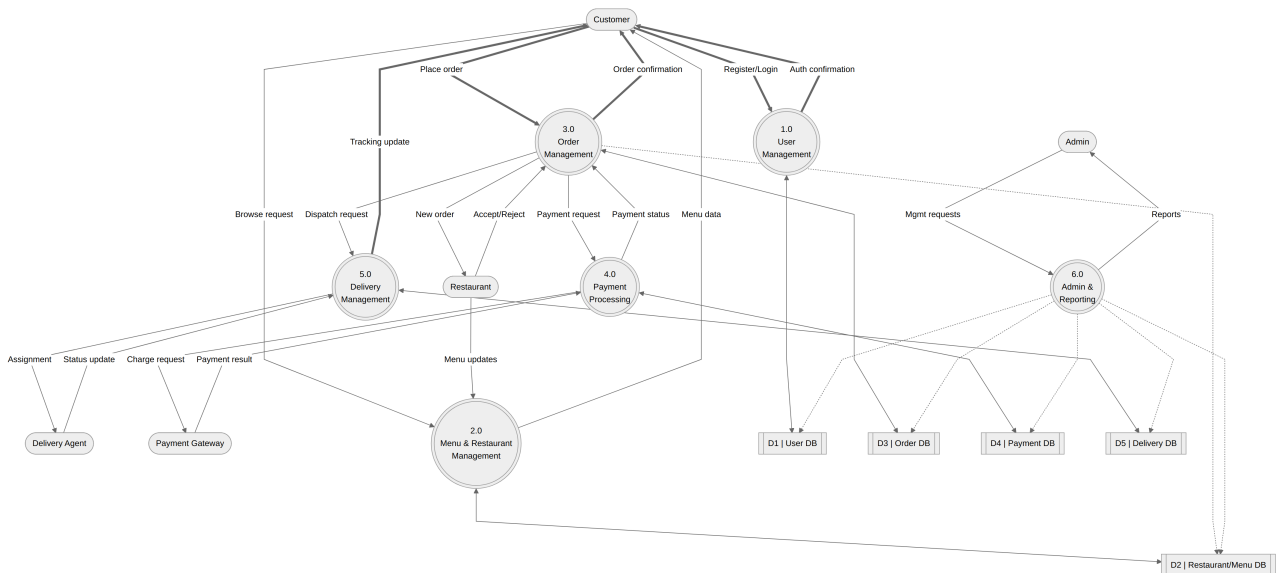


Figure 2: Level 1 DFD

Key Points

- **1.0 User Management** handles registration/login against the User DB (D1).
- **2.0 Menu & Restaurant Management** maintains restaurant and menu data (D2).
- **3.0 Order Management** is the core process, creating and updating orders (D3) and coordinating with Payment (4.0) and Delivery (5.0).
- **4.0 Payment Processing** talks to the external Payment Gateway and logs transactions (D4).
- **5.0 Delivery Management** assigns agents and tracks deliveries (D5).
- **6.0 Admin & Reporting** reads across all data stores to generate reports for the Admin.

3. Level 2 DFD — Order Management (3.0) Expanded

Zooms into process 3.0 from the Level 1 diagram, breaking Order Management down into six detailed sub-processes.

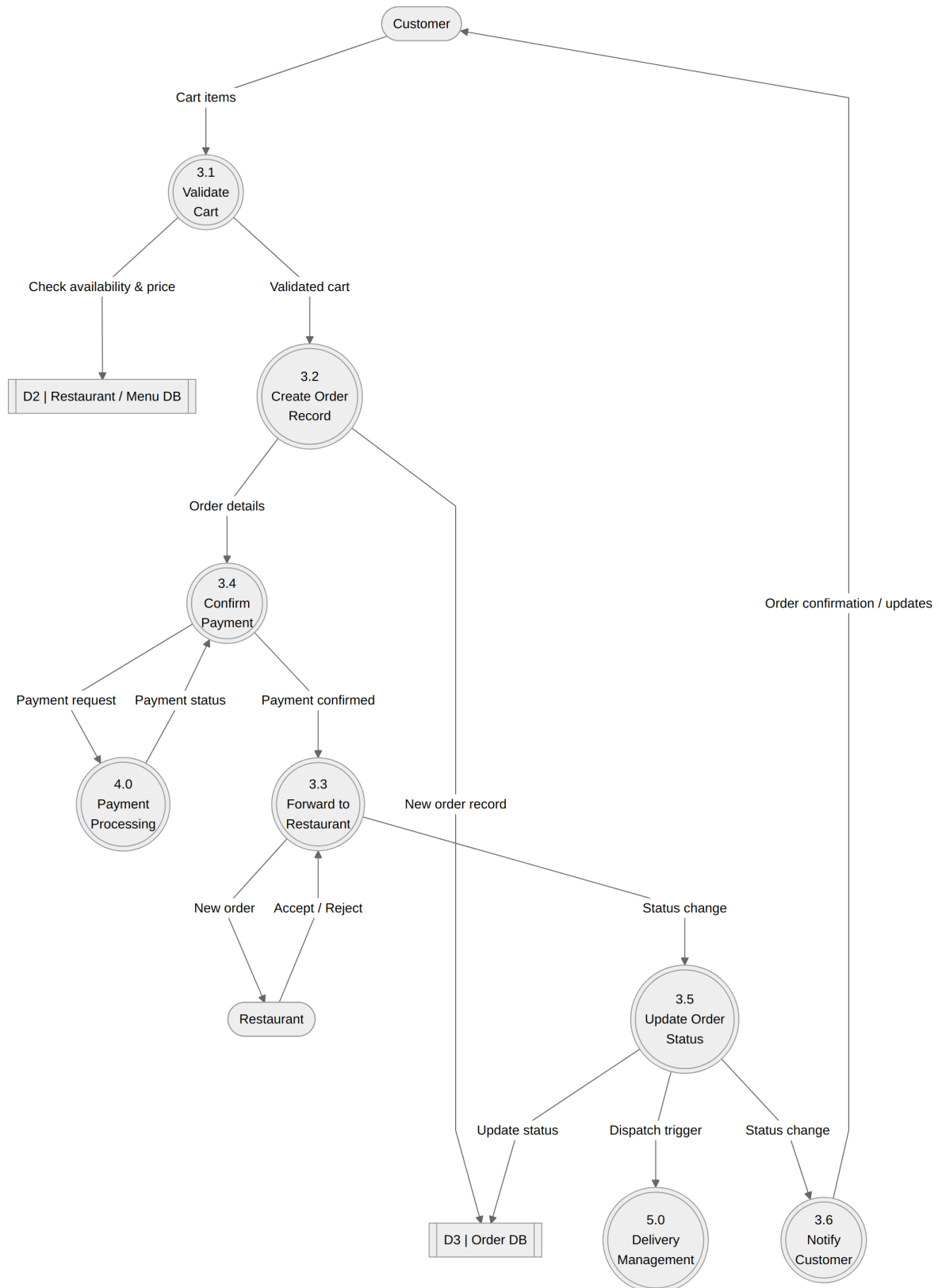


Figure 3: Level 2 DFD — Order Management (3.0) Expanded

Key Points

- **3.1 Validate Cart** checks item availability/pricing against the Menu DB before proceeding.
- **3.2 Create Order Record** persists the new order to the Order DB (D3).
- **3.3 Forward to Restaurant** sends the order for acceptance and receives the decision.
- **3.4 Confirm Payment** hands off to the Payment Processing process (4.0).
- **3.5 Update Order Status** updates D3 and triggers dispatch via Delivery Management (5.0).
- **3.6 Notify Customer** sends confirmation and status updates back to the customer.